

VIPR CoolDawgs App

A User-Centric Application for Canine Heat Stress Management



App Design Team

Nivedha Natarajan, Kristian Pitshugin, Bhargav
Gudapati, Sydney Staggs, Nikita Neeli

Faculty Mentors: Dr. Bradley Barnes,
Dr. Lakshmish Ramaswamy



UNIVERSITY OF
GEORGIA

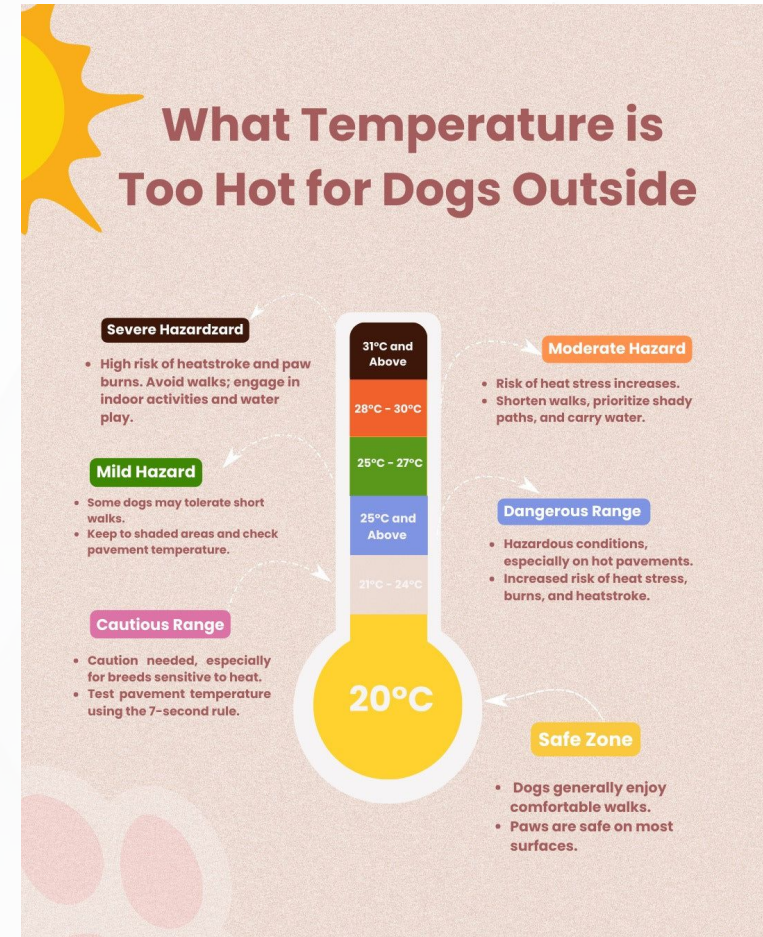


Background + Motivation



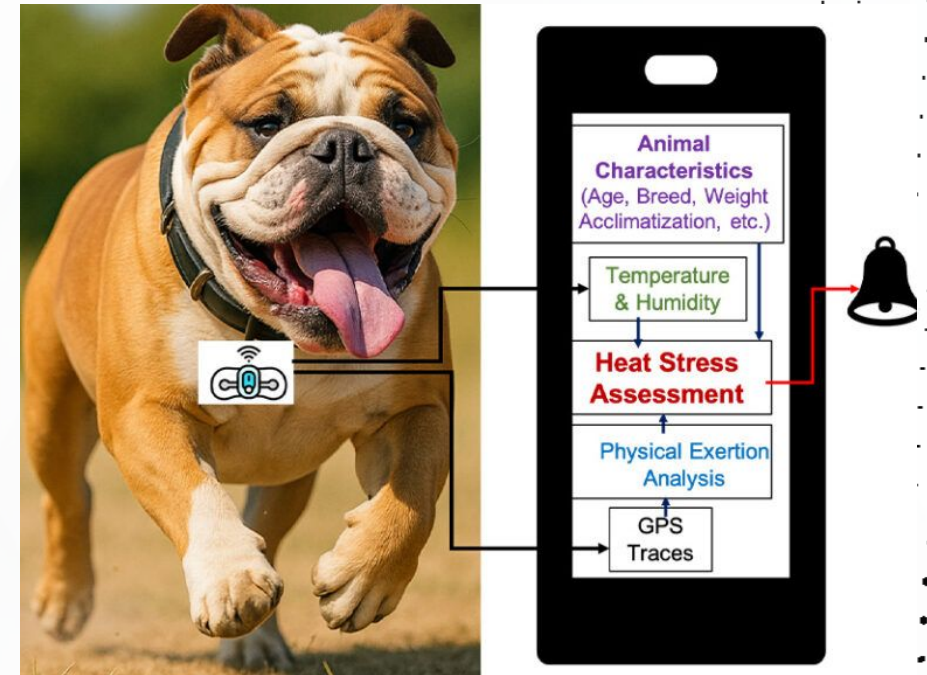
Motivation

- Lower height can lead to greater heat stress
- Better **communication** between owners and their dogs
- Encouraging **timely action** by owners
- **User-friendly** and insightful system to bridge the gap



Existing System + Improvements

- Built with inspiration from existing app made by the team in a previous semester
 - Receives temperature readings from a Kestrel device
 - Displays a graph of the readings, along with a prototype of push notifications to receive alerts
- **Goal:** Provide a full user experience, with pet management, user management, helpful data visualization with graphs, and a user-friendly UI
- Connection to a nonrelational database
 - Support time series data along with the data associated with pet and user management



Features + Design



Design Analysis

- **User-friendly Display**
 - Showcase necessary data to users, making the app accessible and intuitive to use
- **Device Management**
 - Supports the monitoring of multiple pets for each user
- **Pet Profile**
 - Individual pet profiles showcasing the pet's data and temperature statistics
- **Data Visualization**
 - Each pet's temperature and health statistics displayed using a graph of aggregated data
- **Real-time Temperature Tracking**
 - Showing pet's temperature in real time, see past 7 days of temperature data
- **Push Notification**
 - Push notifications when temperature is beyond threshold for pet

User Interface Design

- When designing the user interface, we wanted to make sure we created a **simple, modern, and visually appealing** app for users
- The user shouldn't have to worry about unnecessary information, so we kept their screens clear of any noise. Everything is displayed while prioritizing **intuitive navigation**
- We also decided to go with these colors to embody the **aesthetic** we were looking for and show **school spirit!**

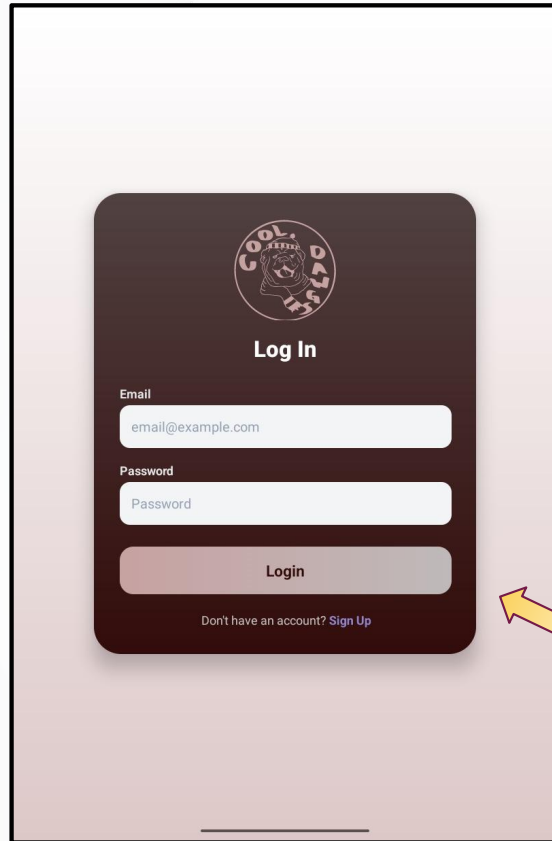


User Interface Design - Logo Design



User Interface Page Breakdown

Login Page

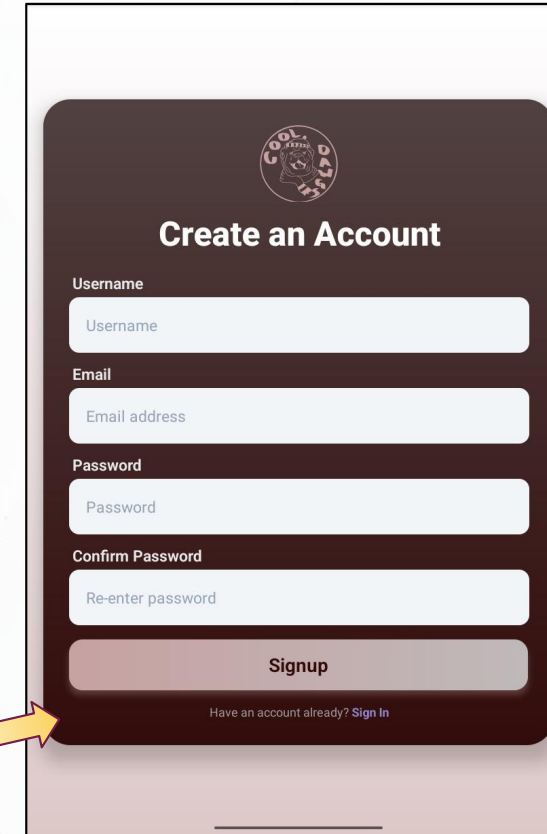


The Login Page features a dark-themed card with a circular logo at the top. Below the logo is the text "Log In". The card contains two input fields: "Email" with the placeholder "email@example.com" and "Password" with the placeholder "Password". A "Login" button is positioned below the password field. At the bottom of the card, there is a link that says "Don't have an account? Sign Up".

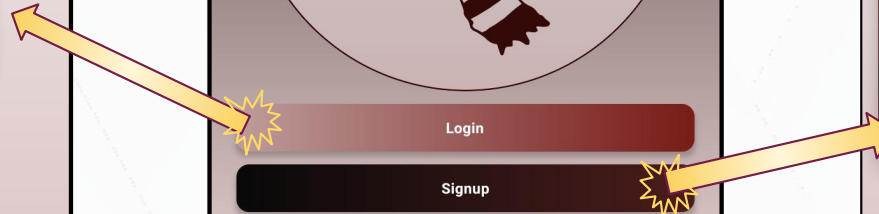
Home Page



Register Page

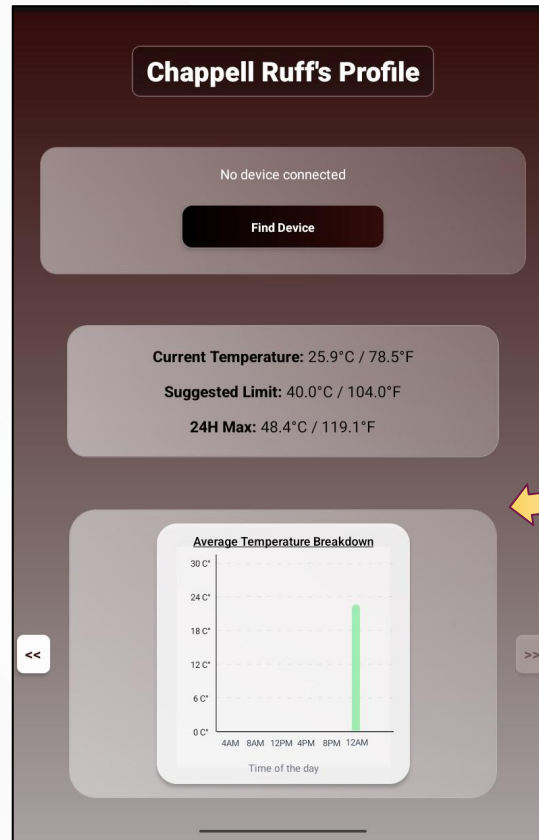


The Register Page features a dark-themed card with a circular logo at the top. Below the logo is the text "Create an Account". The card contains four input fields: "Username" with the placeholder "Username", "Email" with the placeholder "Email address", "Password" with the placeholder "Password", and "Confirm Password" with the placeholder "Re-enter password". A "Signup" button is positioned below the confirm password field. At the bottom of the card, there is a link that says "Have an account already? Sign In".

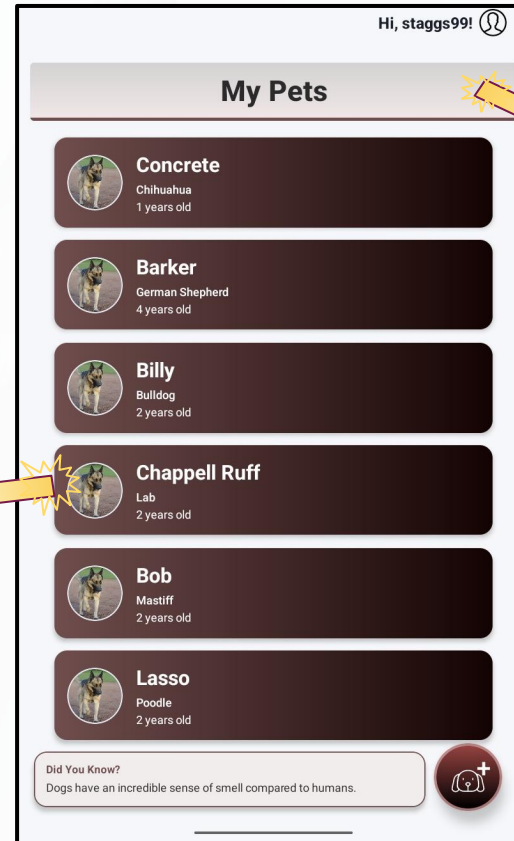


User Interface Page Breakdown

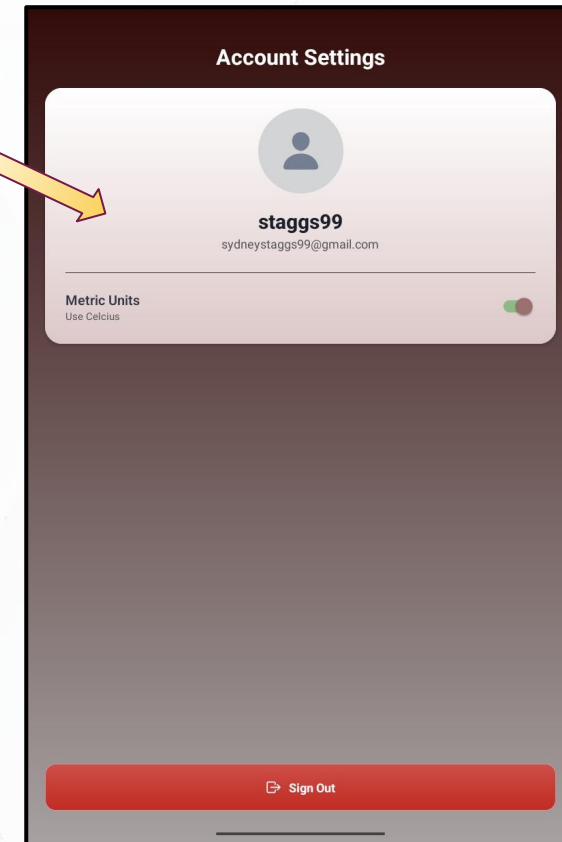
Pet Profile Page



Pet Page

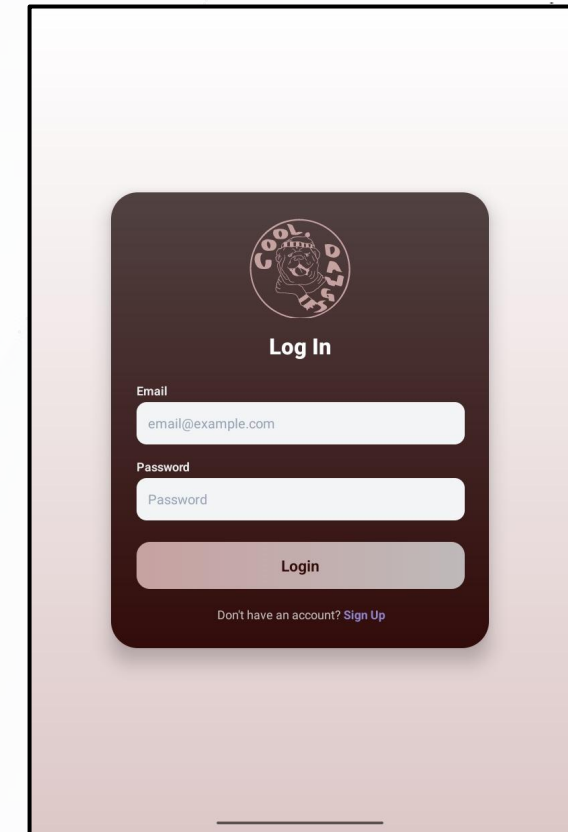


Account Page



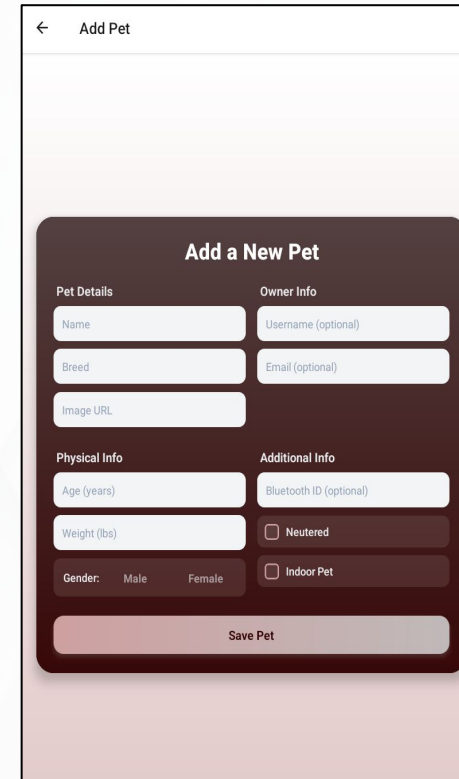
Authentication and Authorization

- **Firestore Authentication:** Email and password
- Encrypted Authentication
- Persistent Authentication
 - **Maintains state** when app is closed and relaunched
- Authentication-based Routing
 - **Not Logged In:** Welcome screen, authentication screens
 - **Logged In:** Home screen, Add Pet Form, Pet Profiles



Pet Management

- **Add Pets**
 - Requests information to determine the heat stress threshold
- **View Pets by User**
 - Retrieved from database
 - Shown on Home Screen
- **Pet Profiles**
 - Individualized profile for each pet

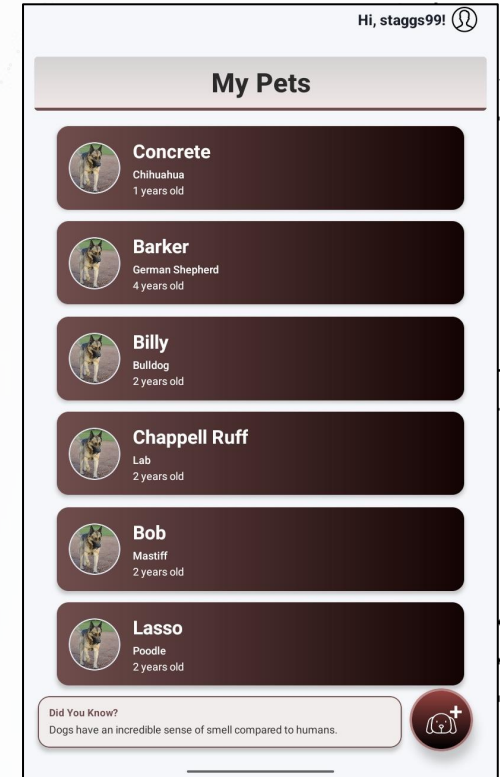


← Add Pet

Add a New Pet

Pet Details	Owner Info
Name	Username (optional)
Breed	Email (optional)
Image URL	

Physical Info	Additional Info
Age (years)	Bluetooth ID (optional)
Weight (lbs)	☐ Neutered
Gender: ☐ Male ☐ Female	☐ Indoor Pet



Hi, stagg99!

My Pets

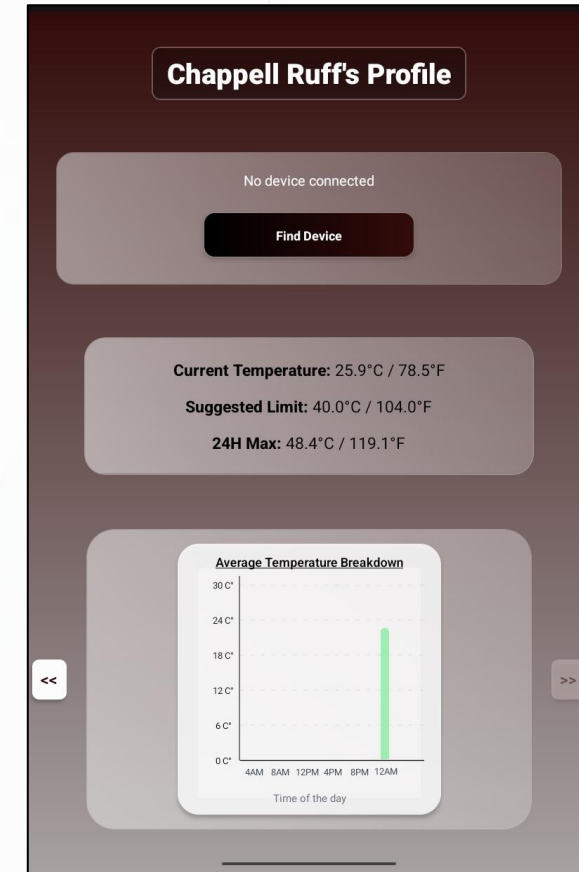
-  **Concrete**
Chihuahua
1 years old
-  **Barker**
German Shepherd
4 years old
-  **Billy**
Bulldog
2 years old
-  **Chappell Ruff**
Lab
2 years old
-  **Bob**
Mastiff
2 years old
-  **Lasso**
Poodle
2 years old

Did You Know?
Dogs have an incredible sense of smell compared to humans.



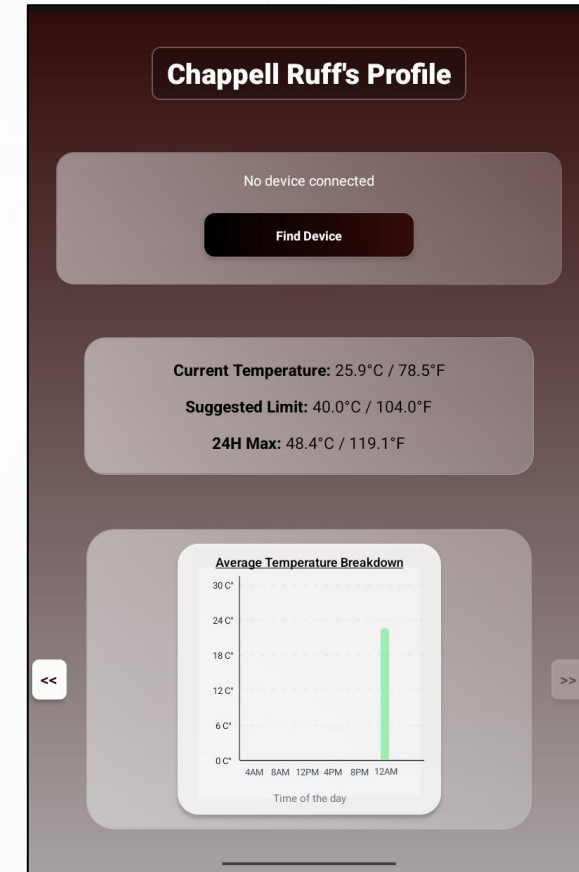
Bluetooth Connection

- Using BLE (Bluetooth Low Energy)
 - **Long range** communication
 - **Low energy** transmission/receival
- Receives data once every 20 seconds
 - Compromise between **battery life** and **data resolution**
- **Easily configurable** from app to connect to any available BLE device



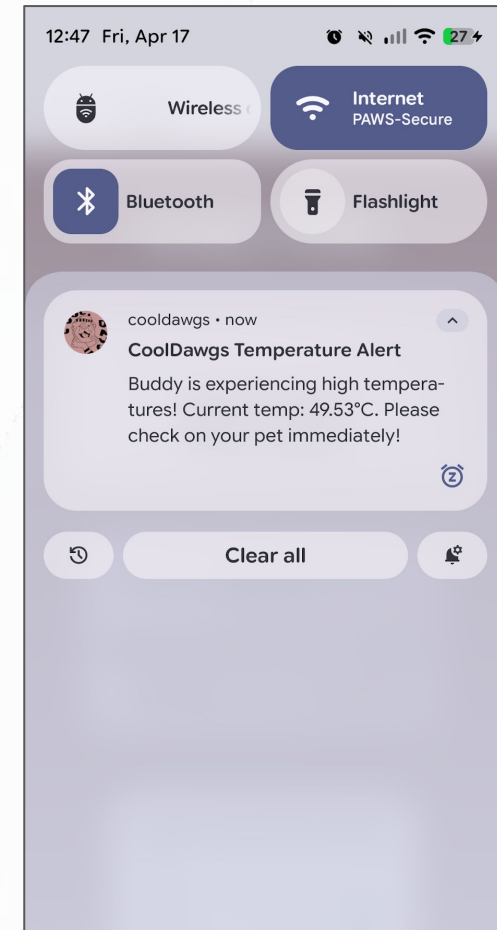
Data Visualization

- Displays the pet's **heat exposure** in a clear chart for quick understanding
- Updates the temperature data at **4-hour intervals** across the last 24 hours
- Next and Previous buttons to view heat exposure data from earlier days across the **past week**
- Fetches **time-series data** from a non-relational Firebase database
- Shows only the most relevant data to improve **clarity** and **accessibility**



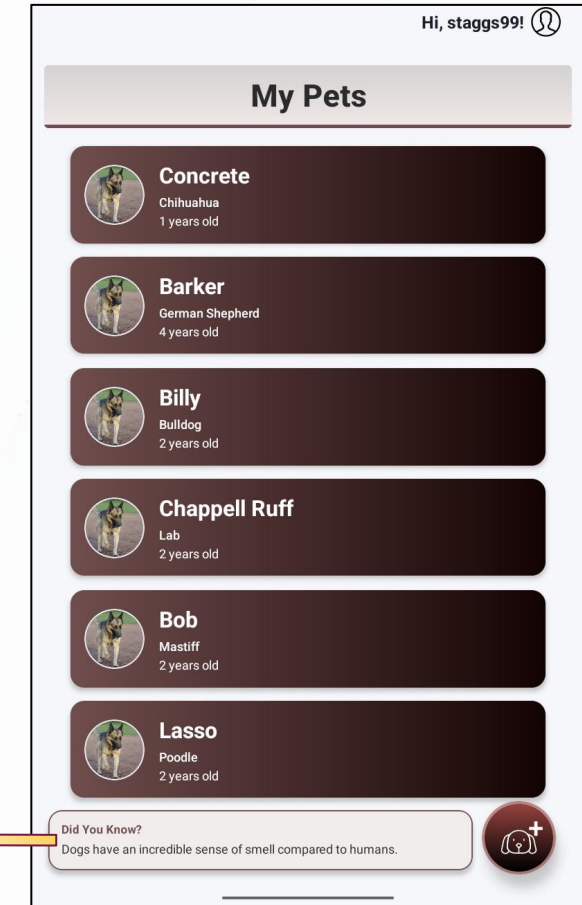
Notification System

- Push notifications when pet is experiencing heat stress
- One alert every **15 minutes** to avoid overuse of warning messages, which could lead to the devaluation of each warning message
- Mentions current temperature and pet name to provide valuable information for **easy tracking**



AI Integration

- In Progress:
 - Integrating **models** developed by the CoolDawgs team to determine heat stress threshold
 - **Computer vision** to detect characteristics and breed of dog
 - AI insights to showcase pet temperature data in an **aggregated, digestible** manner
- AI generated **dog facts** displayed for the user



Future Work



Future Work

- Train a **small language model** (SLM) to reduce dependency on larger external LLMs
- Add a dedicated **AI analytics** interface based on **historical pet data**
- Provide **emergency guidance** for signs of overheating or heat stress
- Include weather and location based **heat warnings**
- Enable **veterinary data sharing** and report export
- Field data collection with **real dogs**

Thank you!

Questions?



UNIVERSITY OF
GEORGIA